LESSON PLAN

WORKSHOP PRACTICES (PRACTICAL) (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	The basic principle of propulsion system used in guided weapons, present day propulsive systems and propellants for rockets and their classifications
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	The basic principle of propulsion system used in guided weapons, present day propulsive systems and propellants for rockets and their classifications. Practical on propulsion system.
5. Specific Objectives	23.1.1.1 Trainee will be able the basic knowledge of air craft armaments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of rocket propulsion model; Safety: PPE mandatory, no loose clothing; Supervision: Instructor monitors ignition and range safety.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the fundamental principle of rocket propulsion?
(b) Name two types of guided weapon propulsion systems.
(c) Define specific impulse in the context of rocketry.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principle of propulsion systems • Present day propulsive systems overview • Rocket propellants types and characteristics • Classification of rocket propellants	L & Ds	PPT, IB	15 min
(b)	• Practical demonstration of propulsion model • Safety protocols for propellant handling • Instructor supervision of live fire simulation • Trainee observation and note taking	L & Ds	PPT, PDF	20 min
(c)	• Review of propellant classifications • Comparison of solid vs liquid systems • Application in guided weapons context • Summary of key propulsion concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between solid and liquid propellants.
(b) Describe the role of thrust vectoring in guided weapons.

12. Home Assignment and Follow Up	Prepare a brief note on the control systems used in guided missiles to be discussed in LP 07.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Guided missile control system, introduction functions and types of control systems (Aerodynamic controls and thrust controls) and classifications of aerodynamic and thrust control.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min